

```
pip install nltk gensim
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
27.9/27.9 MB 63.2 MB/s eta 0:00:00

Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import nltk
from nltk.corpus import brown
from gensim.models import Word2Vec

# Download corpus (only first time)
nltk.download('brown')

# Load first 5000 sentences
sentences = brown.sents()[1:5000]

# Convert words to lowercase
sentences = [[word.lower() for word in sent] for sent in sentences]

# Query words
query_words = ["money", "government", "student", "computer"]

# Different hyperparameter settings
vector_sizes = [50, 100]
window_sizes = [2, 5]

# Train models for different settings
for vec_size in vector_sizes:
    for window in window_sizes:

        print("\n=====")
        print("Vector Size:", vec_size, "| Window Size:", window)
        print("=====")

        # Train Skip-gram model
        model = Word2Vec(
            sentences,
            vector_size=vec_size,
            window=window,
            min_count=2,
            workers=4,
            sg=1 # Skip-gram
        )

        # Vocabulary embeddings
        word_vectors = model.wv

        print("\nVocabulary Size:", len(word_vectors))

        # Find similar words
        for word in query_words:
            if word in word_vectors:
                print(f"\nTop 8 similar words for '{word}':")

                similar_words = word_vectors.most_similar(word, topn=8)

                for w, score in similar_words:
                    print(w, ":", round(score, 3))
            else:
                print(f"\n'{word}' not in vocabulary")
```

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Unzipping corpora/brown.zip.
```

```
=====
Vector Size: 50 | Window Size: 2
=====

Vocabulary Size: 6586
```

```
Top 8 similar words for 'money':  
left : 0.997  
effect : 0.997  
start : 0.997  
history : 0.997  
use : 0.997  
big : 0.997  
chicago : 0.996  
off : 0.996
```

```
Top 8 similar words for 'government':  
meeting : 0.994  
most : 0.994  
back : 0.993  
member : 0.993  
vote : 0.993  
same : 0.992  
past : 0.992  
work : 0.991
```

```
Top 8 similar words for 'student':  
wages : 0.997  
western : 0.997  
population : 0.997  
race : 0.997  
year's : 0.997  
pair : 0.997  
persons : 0.997  
questions : 0.997
```

```
'computer' not in vocabulary
```

```
=====  
Vector Size: 50 | Window Size: 5  
=====
```

```
Vocabulary Size: 6586
```

```
Top 8 similar words for 'money':  
great : 0.995  
field : 0.994  
half : 0.994  
use : 0.994  
single : 0.994  
hospital : 0.994  
turn : 0.993  
stock : 0.993
```

```
Top 8 similar words for 'government':
```